

Quantifying the Impact of Credit Risk Crisis in Indian Public Sector Bank on Sovereign Credit

Quality:

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Abstract: Indian public sector banks are undergoing though a credit risk crisis. This is evident from their current level of Non-Performing Loans. In addition, as per external ratings, all the Indian PSBs are worse than a median quality rated banks. In most economies, Sovereign is assumed to be a guarantor (explicit or implicit) of its large financial institutions or public sector banks. Given the role of sovereign as guarantor, the debt of the banks would move to sovereign balance sheet during periods of banking stress. Hence, in an economy where public sector banks have low credit quality, the effective credit quality of the sovereign would be worse than its standalone rating corresponding only to foreign currency (FC) debt. We have tried to quantify a lower bound of increase in credit risk of Indian sovereign due to low credit quality of its implicitly guaranteed banks. **Under very liberal assumptions, the effective credit quality of Indian sovereign would be around 9 times worse than its credit quality implied by sovereign FC rating. This is primarily due to the low credit quality of its PSBs. If all the PSBs were like a median rated bank, the increase would not be more than 5 times.** Assuming high level of interdependence within PSBs or correlation between PSBs and sovereign would make the differential even more higher.

Table of Contents

Table of Contents	2
1 credit risk in indian public sector banks.....	4
2 METHODOLOGY.....	6
2.1 asset growth process of individual obligors	7
2.2 correlation structure of asset growth process	8
2.3 standalone pd	9
2.4 effective sovereign pd.....	9
2.5 methodology to quantify the spillover impact	10
2.5.1 Derivation/Assumption of Inputs.....	10
2.5.2 Monte Carlo Simulation Algorithm.....	16
3 results.....	18
4 Appendix-A : sovereign pd	19
4.1 Data & Estimation of generator matrix and regeneration of 1 year transition matrix ...	20
4.2 interpolation at the outlook level.....	21
5 Appendix-B : bank pd	23
6 Appendix-C simulation results	24

The objective of this paper is to estimate the additional credit cost of Indian government (GOI) as an implicit guarantor of public sector banks (PSB). The credit cost is estimated as an increase in sovereign credit risk rating (vis-à-vis its standalone credit risk level corresponding to FC debt) on account of increased liability of Indian government in scenarios of distress in PSBs. This is a very relevant issue now due to the prevailing credit risk crisis in PSBs.

Like all other sovereigns, India has an external rating which corresponds to its repayment capacity for foreign currency (FC) debt. As per Moody's latest rating, India's FC rating is Baa3. It is to be noted that this is the lowest investment grade. The rating is primarily related to foreign currency sovereign debt. The Moody's sovereign rating methodology does consider banking sector strength as a rating input¹. However, this does not explicitly consider the sovereign role of implicit guarantor of PSB. Banking sector strength is just used as a proxy for event risk. This is also evident from the fact that Moody's bank rating methodology explicitly considers sovereign support to adjust the standalone rating of banks². Hence, the sovereign rating does not explicitly consider the increase in risk of sovereign as a guarantor of PSBs.

In this paper, we argue that when we consider the role of GOI as an implicit guarantor of PSBs, the real credit risk of India would be significantly different than what is implied from its current rating. Our main argument is that in years of banking distress (one or many banks in stress), the implied debt levels of GOI will increase. In addition to sovereign debt, immediate debts of distressed PSBs would also need to be addressed by GOI. In cases of multiple debt, payment priority would anyway be an issue of seniority of debt structure. In most likelihood, sovereign debt would be having a higher seniority. Nevertheless, increased likelihood of such scenarios in all possibilities is an issue of credit risk.

This issue (i.e. increased credit risk due to implicit guarantee of PSB) would be not a major or significant concern for all sovereigns. The issue becomes prominent for those economies where banks have a high level of credit risk or high level of intra-correlations or both. In case of India, almost all PSB banks are worse than a median level globally rated banks. This is not only evident from their external rating but even by the levels of current NPA. We have not analyzed the level of intra correlation among the banks. However, the simultaneous increase in NPAs does indicate towards some presence of intra correlation among the PSBs.

The plan of the paper is as follows. In section (1), we give a brief overview of current credit risk levels of Indian PSBs. In section (2), we present our methodology. Section (3)

¹ Rating methodology, Sovereign Bond Rating, Moody's Investor Service, December 22, 2016

² Rating methodology, Banks, Moody's Investor Service, January, 2016

present a summary of our results. In section 4& 5, we have explained the derivation of sovereign rating to PD mapping and bank rating to PD mapping respectively. Section 6 presents detailed results for all choices of inputs and assumption.

1 CREDIT RISK IN INDIAN PUBLIC SECTOR BANKS

The level of increased risk within Indian banking system can be seen by the following two metrics:

- 1) Non-Performing Loan Data
- 2) External Ratings

A popular banking risk metric being used in common discussion is non-performing loan (NPL) as % of total gross loan. This has increased in recent years. This is shown in table “India NPL Trend”.

Table: India NPL Trend: Source: World Development Indicators, 30 June 2017

year	Bank nonperforming loans to total gross loans (%) ⁱ
2011	2.67
2012	3.37
2013	4.03
2014	4.35
2015	5.88
2016	9.19

Within 110 countries whose NPL is reported for 2016, Indian ranks at 81. Only countries like Russia, Pakistan, Uganda and some smaller countries are below it.

In terms of External ratings, almost all the PSB are worse than a median rated bank in terms of standalone credit assessment. As per the Moody sovereign rating methodology, the standalone credit quality is expressed by BCA (baseline credit adjustment) denominated in lower cases. As per the recent bank rating data³, the cumulative distribution of banks by BCA is given in the table titled “**BCA Cumulative Distribution**”. Among all the rated banks, median rating is ba1. Within Indian PSBs, the best BCA is for SBI (=ba1). Rest of the banks generally fall at the 70th percentile. This is evident from table titled “**Indian Bank ratings**”. The data is from the same moody website.

³Moody website. <https://www.moodys.com/newsandevents/topics/bank-methodology-ratings-and-research/-/007031/007031/-/1/0/-/0/-/1/-/1/-/en/global/pdf/-/rra>. The file is “Average Bank Ratings by Country (Excel Data)”. The data is given on the tab “Issuer-Level Ratings”.

Table: BCA Cumulative Distribution

BCA	Number of Rated Countries	percentile
aaa	0	0.00%
aa1	0	0.00%
aa2	0	0.00%
aa3	2	0.21%
a1	21	2.47%
a2	43	7.08%
a3	71	14.70%
baa1	86	23.93%
baa2	115	36.27%
baa3	98	46.78%
ba1	70	54.29%
ba2	84	63.30%
ba3	68	70.60%
b1	72	78.33%
b2	55	84.23%
b3	76	92.38%
caa1	39	96.57%
caa2	15	98.18%
caa3	7	98.93%
ca	10	100.00%
c	0	100.00%

Table: Indian Bank Standalone Moody Ratings

Indian PSBs	BCA
Bank of Baroda	ba2
Bank of India	ba3
Canara Bank	ba3
Central Bank of India	b3
IDBI Bank Ltd	caa1
Indian Overseas Bank	b3
Oriental Bank of Commerce	ba3
Punjab National Bank	ba3
State Bank of India	ba1
Syndicate Bank	ba2
Union Bank of India	ba3

2 METHODOLOGY

We have used the stochastic asset growth – fixed default barrier framework (Merton Model) to analyze the impact of implicit guarantees. In this methodology, an obligor has a stochastic asset growth process. At the end of the assessment horizon, the obligor defaults if its terminal asset falls below the fixed debt level.

The above framework can be easily applied to analyze the level of spillover risk from a set of obligors (group of PSB) to a single obligor (sovereign). In case the sovereign is an guarantor, the concept of fixed debt barrier does not hold. In case of any of the bank defaults, the sovereign is expected to pay for the bank debt. In other words, effective sovereign debt at the end of the horizon is the sum total of following:

- 1) Sovereign own debt
- 2) Debt of defaulted banks

As a result, there would be cases when sum of sovereign own debt plus debt of stressed banks exceeds the terminal asset of the sovereign. We call the likelihood of such events as “**Effective Sovereign PD**”. This PD reflects not only the repayment capacity of the sovereign for its own debt but to even take care of the debt of the stressed banks. The repayment capacity of the sovereign with respects to its own debt is termed as “**Standalone PD**” in this paper. The ratio of effective sovereign PD to standalone PD is termed as “**spillover impact**” in this paper.

The concept of sovereign rating and PD is primarily related to its foreign currency (FC) debt. This is also the concept of sovereign PD used in this paper. The FC debt would obviously be a senior debt of the sovereign vis-à-vis the debt of the stressed banks. Hence it is more likely that even in cases where sum of sovereign FC debt and debt of stressed banks falls below the terminal asset of the sovereign, sovereign may chose to ignore bank debts and repay its FC debt. Nonetheless, such scenarios would be cases of sovereign risk.

The difference between Effective Sovereign PD and standalone PD will depend upon the following 4 factors:

Difference in level of standalone PD of the Sovereign and the Banks: In case, banks have high PD vis-à-vis sovereign PD, event of joint stress would be more likely

Correlation between banks and sovereign: High level of correlation of banks with sovereign will imply higher joint likelihood of defaults of banks and low asset growth of sovereign.

Correlation between banks: a group of banks with high level of correlation among themselves will result in more than one defaults if even one of the bank is stressed.

Size of the PSBs vis-à-vis Sovereigns: The impact of all the above three issues would increase if the group of PSB are of high size vis-à-vis the sovereign.

In nutshell, the spillover impact would be high in an economy where high size but low credit quality PSB banks are not only highly correlated with sovereign but also among themselves.

The quantitative description of the above framework is discussed below. The discussion is arranged in the following sequence:

- 1) Asset Growth Process of Individual Obligors
- 2) Correlation Structure of Asset Growth Process
- 3) Standalone PD
- 4) Effective PD
- 5) Methodology to quantify the spillover impact

2.1 ASSET GROWTH PROCESS OF INDIVIDUAL OBLIGORS

Consider an economy with one sovereign (i.e. S) and number of PSBs (PSB_i , where $i = 1, 2, \dots, N$) with implicit guarantees from the sovereign. Any obligor (sovereign or bank) will default if its relevant asset⁴ at the end of horizon⁵ falls below its debt⁶. The asset generation process from start of the year to its end point is given by the Merton model. The asset generation process for the sovereign (s) is given below:

$$A_s(T) = A_s(0) \cdot \exp \left[\left(\mu_s - \frac{\sigma_s^2}{2} \right) T + (\sigma_s \sqrt{T}) w_s \right] \quad (1)$$

In the above equation, $A_s(0), A_s(T)$ are the value of asset at the starting point and end point of the horizon. μ_s, σ_s denote mean and standard deviation of the asset return of the obligor. w_s denotes the stochastic element of the asset generation process and is a random realization from standard normal distribution.

Similarly, the asset generation process for the PSBs is as follows:

⁴ This is not the book value of asset but market value of the asset

⁵ This is the payment point.

⁶ This is not the entire debt of the obligor but only limited to the required payment at that point.

$$A_i(T) = A_i(0) \cdot \exp \left[\left(\mu_i - \frac{\sigma_i^2}{2} \right) T + (\sigma_i \sqrt{T}) w_i \right] \quad (2)$$

2.2 CORRELATION STRUCTURE OF ASSET GROWTH PROCESS

The asset generation process of PSBs is expected to be correlated with the asset generation process of sovereign. In addition, banks could be correlated among themselves due to some common risk or individual dependencies among themselves. These two level of dependencies can be expressed by the following expression:

$$w_i = \rho_{S,i} \cdot w_s + \left(\sqrt{1 - \rho_{S,i}^2} \right) \left[\rho_{B,i} \cdot w_B + \left(\sqrt{1 - \rho_{B,i}^2} \right) \cdot w_{i_idiot} \right] \quad (3)$$

Where w_s denote the stochastic element in the asset generation process of the sovereign (equation (1)). w_B is a proxy for general risk in the banking sector. In addition, w_B is independent of w_s . Both of these two factors follow $N(0,1)$. $\rho_{s,i}, \rho_{B,i}$ are the loadings of bank i on w_s, w_B respectively and are in the range $[0,1]$. w_{i_idiot} is a factor specific to bank i and is independent of the other two risk factors and follows $N(0,1)$. The above correlation structure has the following properties:

- w_i follows $N(0,1)$ (as required for equation 2)
- Correlation between any two banks equals: $\rho_{S,i}^2 + (1 - \rho_{S,i}^2) \cdot \rho_{B,i}^2$. Hence, even if there is no correlation with sovereign, correlation between any two banks equals $\rho_{B,i}^2$

The above structure would require estimation of separate parameters for each of the customers and hence the computation levels would be too much. Hence, we have used a relatively simplified process (i.e. same factor loadings for all banks). Use of same factor loadings for all banks may not be exactly correct. However, given the fact that our target PSBs are based in a single country, the implied error would be less. The simplified correlation structure is as follows:

$$w_i = \rho_S \cdot w_s + \left(\sqrt{1 - \rho_S^2} \right) \left[\rho_B \cdot w_B + \left(\sqrt{1 - \rho_B^2} \right) w_{i_idiot} \right] \quad (4)$$

2.3 STANDALONE PD

Given the above discussion of asset growth and a given fixed debt, the standalone probability of default (for 1 year horizon) for sovereign and PSB banks is given by the following equations

$$PD_s = pr \left[w_s < \frac{\ln\left(\frac{D_s}{A_s(0)}\right) - \mu_s + \frac{\sigma_s^2}{2}}{\sigma_s} \right] = N \left[\frac{\ln\left(\frac{D_s}{A_s(0)}\right) - \mu_s + \frac{\sigma_s^2}{2}}{\sigma_s} \right] \quad (5)$$

$$PD_i = pr \left[w_i < \frac{\ln\left(\frac{D_i}{A_i(0)}\right) - \mu_i + \frac{\sigma_i^2}{2}}{\sigma_i} \right] = N \left[\frac{\ln\left(\frac{D_i}{A_i(0)}\right) - \mu_i + \frac{\sigma_i^2}{2}}{\sigma_i} \right] \quad (6)$$

where $N[\cdot]$ is the cumulative probability of standard normal distribution and i denotes PSBs

2.4 EFFECTIVE SOVEREIGN PD

In case of sovereign being a guarantor of PSB (explicit or implicit), the following 5 cases are possible:

Case	Sovereign	Banks	Repayment of Combined Debt	Sovereign Default
1	Terminal Asset > FC Debt	Terminal Asset > Debt	Not Applicable	NO
2	Terminal Asset > FC Debt	Terminal Asset < Debt for one or many banks	Terminal Asset of Sovereign > FC Debt + total debt of the stressed banks	NO
3	Terminal Asset > FC Debt	Terminal Asset < Debt for one or many banks	Terminal Asset of Sovereign < FC Debt + total debt of the stressed banks	YES
4	Terminal Asset < FC Debt	Terminal Asset > Debt	Not Relevant	YES
5	Terminal Asset < FC Debt	Terminal Asset < Debt for one or many banks	Not Relevant	YES

In case, sovereign was not a guarantor, its PD (i.e. **Standalone PD**) is captured by 4th and 5th Case. Given the sovereign role as a guarantor, **Effective PD** is the combined probability of cases 3, 4 and 5.

The spillover impact has been quantified by using a Monet Carlo Simulation approach. In order to run the simulation approach, following inputs would be required:

- We have not computed many of the above inputs. Our objective is to approximate the lowest level of spillover impact. Hence, based upon the dependency of spillover impact on the inputs, we have assumed corresponding values. This has been complemented by a sensitivity analysis.

- 1) Derivation/Assumption of Inputs
- 2) Monte Carlo simulation Algorithm

2.5.1.1 Initial Asset of the Sovereign and Bank

The choice of initial asset value for banks have been also set in relativity. We assume that the entire market value of total bank assets would be a proportion of initial asset of sovereign (i.e. 1). For example, one can assume that the sovereign asset would be 10 times to that of the total asset of the PSBs. In other words, total asset of the PSB would be equal to .1 (i.e. 1/10 of sovereign asset). Please note that this generalization is not intended to suggest that general sovereign assets are 10 times of total PSB assets. The asset referred here are simply corresponding to those mentioned in the Merton model.

10

[/rra](#). The file is “Average Bank Ratings by Country (Excel Data)”. The data is given on the tab “Issuer-Level Ratings”. The data is produced in the table titled “**Rated Indian Banks Assets**”.

The distribution of book value of asset may or may not be identical to a distribution implied by market value of assets. An independent research paper⁷ has attempted to estimate the market value of Indian PSB (except for Central Bank of India). This is shown in column 4 of the table titled “**Rated Indian Banks Assets**”. A comparison of distribution implied by market value of assets (column 5) is almost identical to the distribution implied by book value of assets (column 6).

Hence, we do not believe that using ratios implied by book value of assets would be unreasonable.

Given the above discussion, the initial asset value of sovereign and banks is set as follows:

- 1) Initial asset value of sovereign equal 1
- 2) Initial asset value of a bank is derived by the following expression:
$$A_i(0) = per_i / sov_bank_ratio$$
 where per_i is as given in column 3 of the table “**Rated Indian Banks Assets**” and sov_bank_ratio is the ratio of initial sovereign asset to total of initial PSB assets
- 3) **We are not sure about the accurate value of sov_bank_ratio . Hence, we have done our analysis with 3 values (10, 4, 2).**

⁷ Market Valuation and Risk Assessment of Indian Banks using Black –Scholes -Merton Model”, Pankaj Sinha and Sakshi Sharma and Kriti Sondhi, Faculty of Management Studies, University of Delhi, 6. June 2013

Table: Rated Indian Bank Assets and Market Value from MPRA paper

1	2	3	4	5	6
PSB	Assets (USD Million)	Percentage of Total	Market Value of Asset (INR Crores) (MPRA Paper)	Percentage (book value) without central bank of India	Percentage (market value) without central bank of India from MPRA Paper
Bank of Baroda	100,966	10.08%	418863	11%	11%
Bank of India	91,756	9.16%	355170	10%	9%
Canara Bank	83,287	8.31%	344315	9%	9%
Central Bank of India	46,024	4.59%	Not available		0%
IDBI Bank Ltd	56,019	5.59%	241078	6%	6%
Indian Overseas Bank	41,305	4.12%	198554	4%	5%
Oriental Bank of Commerce	35,694	3.56%	160208	4%	4%
Punjab National Bank	100,340	10.01%	427192	10%	11%
State Bank of India	339,443	33.88%	1291724	36%	34%
Syndicate Bank	46,297	4.62%	164238	5%	4%
Union Bank of India	60,882	6.08%	218269	6%	6%
Total	1,002,012		3,819,611		

2.5.1.2 Mean Return & Volatility of Sovereign & Banks

2.5.1.2.1 Asset Return Volatility

We have set asset return volatility of PSBs at 2%. This as per estimated in the above mentioned MPRA paper. As per the paper estimate, volatility of asset return of public sector banks is around 2%. It is reasonable to assume that the asset return volatility of sovereign would not exceed that of PSBs.

Hence asset return volatility is set at 2% for both sovereign and the PSBs

2.5.1.2.2 Mean Return

In case of sovereign, this is set at 6%⁸. In case of PSBs, we have used values (4%, 6%, 8%)

⁸ annualized average of 90 day T-bill rate as given on RBI website

2.5.1.3 Debt/Default Barrier for sovereign and banks

Given equation (5) & (6) in the section on standalone PD, default barrier can be estimated by the following equation for sovereign and PSB;

$$D_s = A_s(0) \cdot \exp \left[N^{-1}(PD_s) * \sigma_s - \frac{\sigma_s^2}{2} + \mu_s \right] \quad (7)$$

$$D_i = A_i(0) \cdot \exp \left[N^{-1}(PD_i) * \sigma_i - \frac{\sigma_i^2}{2} + \mu_i \right] \quad (8)$$

In order to derive default barrier using the above equations, we need initial asset, mean return, asset return volatility and PD. Except for the PD, the rest have been discussed earlier.

We have calculated the PDs used in the above equation by using moody ratings for sovereign and financial institutions. The estimation and estimate for the sovereign and banks are discussed below:

2.5.1.3.1 Sovereign PD

In case of sovereign, we have estimated one year PD for each FC rating. As per the latest Moody Rating of India (Baa3), the sovereign PD for our computation equals .06%. The rating to PD mapping is given in the table titled “Sovereign Rating to PD Mapping”. The derivation is explained in Appendix-A

Table: Sovereign Rating to PD Mapping

Sovereign FC Rating	One Year PD
Aaa	0.00003%
Aa1	0.00010%
Aa2	0.00036%
Aa3	0.00083%
A1	0.00191%
A2	0.00439%
A3	0.00736%
Baa1	0.01233%
Baa2	0.02068%
Baa3	0.06289%
Ba1	0.19112%
Ba2	0.57924%
Ba3	0.96874%
B1	1.61588%
B2	2.68362%
B3	4.04266%
Caa1	6.04719%
Caa2	8.95291%
Caa3	13.06077%
Ca	18.66700%
C	25.96107%

2.5.1.3.2 Bank PD

In case of banks, there are many layers of ratings. As per the moody sovereign rating methodology document, BCA is the standalone risk assessment of the bank. The final counterparty rating reflects government support. The standalone rating of the bank is termed as Baseline Credit Assessment (i.e. **BCA**). For the sake of differentiation, BCA is written in lower case letter. The latest Moody rating for Indian PSB is given in the table below:

Table: Indian Banks Ratings

public_sector_bank	BCA	Counterparty Rating
Bank of Baroda	ba2	Baa3
Bank of India	ba3	Baa3
Canara Bank	ba3	Baa3
Central Bank of India	b3	Ba1
IDBI Bank Ltd	caa1	Ba2
Indian Overseas Bank	b3	Ba1
Oriental Bank of Commerce	ba3	Baa3
Punjab National Bank	ba3	Baa3
State Bank of India	ba1	Baa3
Syndicate Bank	ba2	Baa3
Union Bank of India	ba3	Baa3

Source: Public Sector Banks – India, Increasing risk to credit profiles as capital shortfall threatens, Moody’s Investor Service, 20 June 2017

It is evident from the above table that due to assumed sovereign support, the counterparty ratings is set up at Baa3 for most PSBs. It is to be noted that India Sovereign FC rating equal Baa3. Given the inclusion of government support in Counterparty rating, these ratings cannot be used in our analysis. What we need is the standalone credit quality of the bank. Hence, our relevant rating is BCA.

In order to calibrate BCA grades to quantitative PDs, we would need actual count of BCA events. BCA events are primarily those cases where banks are in a stressed situation and require help/capital injection from Sovereign/Central Bank. There is no such literature to count these events at BCA grade level.

Moody reports default data at counterparty rating level for banks. The most recent publication (in 2010) reported a migration matrix for 1 year. We have used that data to calibrate counterparty rating to 1 year PD. In order to get BCA level PD, we have used the rank mapping approach. In nutshell, if a counterparty rating of Baa3 is assigned a PD of x%, then corresponding BCA rating of same rank (i.e. baa3) also gets the same PD.

The derivation of counterparty rating to 1 year PD is given in Appendix B. The final mapping table to be used in our analysis titled “**Bank PD**” is given below:

Table: Bank PD

1	2	3	4 (=2)
Counterparty Rating	PD	Banks BCA	
Aaa	0.00222%	aaa	0.0022%
Aa1	0.00971%	aa1	0.0097%
Aa2	0.04240%	aa2	0.0424%
Aa3	0.05755%	aa3	0.0575%
A1	0.07811%	a1	0.0781%
A2	0.10600%	a2	0.1060%
A3	0.15498%	a3	0.1550%
Baa1	0.22653%	baa1	0.2265%
Baa2	0.33101%	baa2	0.3310%
Baa3	0.53541%	baa3	0.5354%
Ba1	0.86494%	ba1	0.8649%
Ba2	1.39444%	ba2	1.3944%
Ba3	2.09474%	ba3	2.0947%
B1	3.13555%	b1	3.1356%
B2	4.66885%	b2	4.6689%
B3	7.27334%	b3	7.2733%
Caa1	11.16065%	caa1	11.1606%
Caa2	16.75025%	caa2	16.7503%
Caa3	24.37133%	caa3	24.3713%
Ca	34.04198%	ca	34.0420%
C	45.25383%	c	45.2538%

Based on above mapping, the one year standalone PD of the Indian PSBs is reported below:

Table: Standalone PD of Indian PSBs

public_sector_bank	BCA	Counterparty Rating	Standalone PD (1 year)
Bank of Baroda	ba2	Baa3	1.39%
Bank of India	ba3	Baa3	2.09%
Canara Bank	ba3	Baa3	2.09%
Central Bank of India	b3	Ba1	7.27%
IDBI Bank Ltd	caa1	Ba2	11.16%
Indian Overseas Bank	b3	Ba1	7.27%
Oriental Bank of Commerce	ba3	Baa3	2.09%
Punjab National Bank	ba3	Baa3	2.09%
State Bank of India	ba1	Baa3	0.86%
Syndicate Bank	ba2	Baa3	1.39%
Union Bank of India	ba3	Baa3	2.09%

2.5.1.4 Factor Loadings

We have used three values for both the exogenous factors (i.e sovereign factor and bank factor). These are 25%, 50% and 75%. Hence we have 9 combinations.

2.5.2 Monte Carlo Simulation Algorithm

As discussed previously, we have made some choices and assumptions for our inputs. To summarize, these are outlined below:

- 1) Initial Asset of the Sovereign to be fixed at 1
- 2) In order to estimate total bank assets, we have assumed 3 scenarios of ratio of sovereign asset to total bank assets (10,4,2)
- 3) Given the derivation of total bank asset, bank level asset is derived by using the distribution of book value assets
- 4) For Sovereign, mean return and volatility is fixed at 6% and 2% respectively
- 5) In case of banks, volatility is fixed at 2%. However, we have decided to use 3 scenarios of return (i.e. 4%, 6% and 8%)
- 6) Sovereign and bank level default barrier is endogenous
- 7) We have decided to have 3 scenarios for factor loading for sovereign and bank factor (i.e. 25%, 50% and 75%). Hence, we have 9 combinations

In summary, for each choice of ratio of sovereign asset to total bank asset, we have 27 scenarios. Hence, in total, we have 81 scenarios.

With the above list of inputs, our simulation methodology is given below:

Step 1: Derivation of Initial Assets: The sovereign initial asset is fixed at 1. Given the choice of ratio to sovereign asset to total bank assets, bank level asset is determined by the following equation: $A_i(0) = per_i / sov_bank_ratio$ where per_i is as given in column 3 of the table “**Rated Indian Banks Assets**” and sov_bank_ratio is the ratio of initial sovereign asset to total of PSB assets

Step 2: Extraction of Sovereign and Bank Risk Factors in a random manner from standard normal distribution. This is done for 1 million times.

Step 3: Extraction of Bank Level Risk Factor: for each extraction, bank level risk factors are extracted by the following expression:
 $w_i = \rho_{S,i} \cdot w_s + (\sqrt{1 - \rho_{S,i}^2}) [\rho_{B,i} \cdot w_B + (\sqrt{1 - \rho_{B,i}^2}) w_i_idiot]$ ⁹. This step is done for each PSB (i.e. $i = 1, 2, 3, 11$)

Step 4: Calculation of terminal asset of sovereign and each bank by the equations:

Sovereign: $A_s(T) = A_s(0) \cdot \exp \left[\left(\mu_s - \frac{\sigma_s^2}{2} \right) T + (\sigma_s \sqrt{T}) w_s \right]$

⁹ This is from equation 3

PSBs: $A_i(T) = A_i(0) \cdot \exp \left[\left(\mu_i - \frac{\sigma_i^2}{2} \right) T + (\sigma_i \sqrt{T}) w_i \right]$

Step 5: Calculation of default threshold for sovereign and each bank by the following equation:

Sovereign: $D_s = A_s(0) \cdot \exp \left[N^{-1}(PD_s) * \sigma_s - \frac{\sigma_s^2}{2} + \mu_s \right]$

PSBs: $D_i = A_i(0) \cdot \exp \left[N^{-1}(PD_i) * \sigma_i - \frac{\sigma_i^2}{2} + \mu_i \right]$

Step 6: Flagging Sovereign & Bank Defaults: For each simulation sovereign and each of the 11 banks are flagged as standalone default if their respective terminal assets are less than their respective default barriers.

Step 7: Calculating effective default barrier for sovereign: For each simulation, the effective default barrier for sovereign is defined by the following equation:

$$effective_default_barrier = D_s + \sum_{i=1}^{11} def_i \cdot D_i,$$

Where $def_i = 1$ implies default of bank i , else 0

Step 8: Calculating effective sovereign default flag: For each simulation, if the terminal asset of the sovereign is less than its effective default barrier (explained in above step), this is flagged as 1 else 0:

Step 9: Estimating Standalone default count & Effective default count for sovereign: Standalone default count is count of simulations flagged as sovereign default in step 6. Effective default count is count of simulations flagged as effective sovereign default in step 8

Step 10: Estimating Spillover Effect: This is defined as ratio of Effective default count to Standalone default count.

3 RESULTS

We have calculated the spillover impact for each of the 27 scenarios given the value of initial sovereign asset to total bank asset ratio. For all the three scenarios of initial sovereign asset to total bank asset ratio, the minimum spillover impact is 9 times. In other words, the standalone pd of sovereign is 6 basis point (i.e. .06%). However, as an implicit guarantor, this becomes 54 basis point (.54%). In terms of mapping to sovereign FC rating, this level of PD would correspond to a rating of Ba2. This is like a two notch downgrade from its existing rating.

The above particular impact corresponds to the following inputs in the simulation:

- 1) **Diversified Banks isolated from sovereign risk:** In this scenario factor loadings on sovereign risk and bank risk equals 25%
- 2) **Small Banking Sector vis-à-vis sovereign** (sovereign is 10 times of total PSBs)

The summarized results is given in the table below. The detailed simulation results are given in Appendix C

Table: Spill Over Impact Summary

Sovereign Asset to Total Bank Asset	Minimum Spillover Ratio	Maximum Spill Over Ratio
10	9	41
4	43	98
2	108	162

A major factor for high spillover impact is standalone credit quality of the bank. Except for SBI, almost all of them are below median rating banks in terms of BCA. If all of them have a credit rating similar to SBI (ba1), the spillover impact will be halved. This is shown in the table below:

Table: Spill Over Impact Summary (assuming every bank has rating equal to SBI)

Sovereign Asset to Total Bank Asset	Mimum Spillover Ratio	Maximum Spill Over Ratio
10	5	21
4	21	39
2	42	66

4 APPENDIX-A : SOVEREIGN PD

Moody regularly publish 1 year transition matrix of its sovereign FC ratings. These are at grade level and hence one can use them for estimation of 1 year grade level PD. Thereafter, these grade level PDs can be interpolated to get outlook level PD.

However, grade level PD estimation by using 1 year transition matrix is not so straightforward due to zero default rate in investment grades. Hence, the outlook level interpolation becomes even more challenging. One possible way to deal with this problem is to regularize the transition matrix to have certain desirable properties. For our computation, we are mainly concerned with the following two properties:

- 1) Positive default rate for each grade
- 2) Monotonicity in default rates

One way to regularize the transition matrix is to use the following algorithm:

- 1) Estimate a generator matrix¹⁰ for 1 year transition matrix
- 2) Use the generator matrix to reproduce the 1 year transition matrix

The one year transition matrix produced by step (2) above will have positive and monotonic default rate for each grade. This can be used for outlook level interpolation.

The details of the above step is discussed in following sequence:

- 1) Data & Estimation of generator matrix and regeneration of 1 year transition matrix
- 2) Interpolation at outlook level

¹⁰ A generator matrix is a matrix with properties such that $\exp(tQ) = M$ where t = number of years, Q is the generator matrix and M is the transition matrix for horizon equal to t

4.1 DATA & ESTIMATION OF GENERATOR MATRIX AND REGENERATION OF 1 YEAR TRANSITION MATRIX

The migration matrix for moody FC rating is tabled below:

Table: Sovereign FC Rating 1 year Migration Data

From/To:	Aaa	Aa	A	Baa	Ba	B	Caa-C	Withdrawn	Defaulted
Aaa	96.90%	2.94%	0.04%	0.09%	0.00%	0.00%	0.00%	0.04%	0.00%
Aa	3.46%	92.91%	2.11%	0.79%	0.12%	0.00%	0.00%	0.60%	0.00%
A	0.00%	4.17%	91.22%	3.25%	1.29%	0.07%	0.00%	0.00%	0.00%
Baa	0.00%	0.00%	6.10%	89.10%	4.24%	0.51%	0.04%	0.00%	0.00%
Ba	0.00%	0.00%	0.00%	8.32%	85.34%	5.33%	0.29%	0.13%	0.58%
B	0.00%	0.00%	0.00%	0.00%	4.76%	88.66%	3.47%	0.41%	2.71%
Caa-C	0.00%	0.00%	0.00%	0.00%	0.12%	17.83%	67.58%	1.50%	12.97%

Source: Exhibit 8: Sovereign Default and Recovery Rates, 1983-2015 (Excel data), Moody Investor Service

The data given in exhibit 8 has been put to following processing

- 1) Each row has been normalized to sum up to 100% by dividing each cell with the sum of row. This has been done because the reported data may not have all the decimals.
- 2) All the columns except for withdrawn has been re-estimated by dividing it by (100%-withdrawal rate (i.e. column Withdrawn)).
- 3) The column withdrawn is removed
- 4) An extra row (i.e. D) is added at the bottom of the table. All the cells of this row (except for one in the last column) are set to zero. The cell in the last column is set to 1

The resulting data has been put to log expansion by using the expression $\sum_{i=1}^{i=101} (-1)^{i+1} \frac{X-I}{i}$.

The output of the log expansion is given in the table titled “generator_v1”. The generator do not have the required properties¹¹. Hence, we need to regularize the generator. The generator has been regularized by Weight adjustment (WA) methodology. The resulting generator matrix is given in table titled “**final generator**”.

The final generator has been expanded to regenerate 1 year transition matrix. The expansion is done by the following:

$$I + \sum_{i=1}^{101} \frac{x^i}{(factorial)i}$$

The regenerated transition matrix is given in the table titled “**Regenerated 1 year transition matrix**”. The matrix has all the desirable properties:

¹¹ Diagonal elements should be negative, off diagonal should be positive, sum of the row should be equal to zero

Table: generator_v1

I	Aaa	Aa	A	Baa	Ba	B	Caa-C	D
Aaa	(0.0317)	0.0309	0.0000	0.0009	(0.0000)	(0.0000)	(0.0000)	0.0000
Aa	0.0366	(0.0686)	0.0228	0.0083	0.0010	(0.0001)	(0.0000)	(0.0000)
A	(0.0008)	0.0452	(0.0936)	0.0352	0.0137	0.0003	(0.0000)	(0.0000)
Baa	0.0000	(0.0015)	0.0679	(0.1189)	0.0482	0.0042	0.0004	(0.0002)
Ba	(0.0000)	0.0001	(0.0033)	0.0958	(0.1613)	0.0609	0.0024	0.0053
B	0.0000	(0.0000)	0.0001	(0.0026)	0.0550	(0.1227)	0.0446	0.0255
Caa-C	(0.0000)	0.0000	(0.0000)	0.0003	(0.0050)	0.2320	(0.3824)	0.1551
D	-	-	-	-	-	-	-	-

Table: final generator

	Aaa	Aa	A	Baa	Ba	B	Caa-C	D
Aaa	(0.0317)	0.0308	0.0000	0.0009	-	-	-	0.0000
Aa	0.0366	(0.0686)	0.0228	0.0083	0.0010	-	-	-
A	-	0.0450	(0.0940)	0.0351	0.0137	0.0003	-	-
Baa	0.0000	-	0.0674	(0.1197)	0.0478	0.0042	0.0004	-
Ba	-	0.0001	-	0.0949	(0.1629)	0.0603	0.0024	0.0053
B	0.0000	-	0.0001	-	0.0545	(0.1240)	0.0441	0.0253
Caa-C	-	0.0000	-	0.0003	-	0.2305	(0.3849)	0.1541
D	-	-	-	-	-	-	-	-

Table: Regenerated 1 year Transition Matrix

	Aaa	Aa	A	Baa	Ba	B	Caa-C	D
Aaa	96.93342%	2.93500%	0.03642%	0.09127%	0.00359%	0.00025%	0.00002%	0.00003%
Aa	3.48086%	93.46936%	2.12617%	0.79692%	0.12074%	0.00527%	0.00032%	0.00036%
A	0.07720%	4.14931%	91.17969%	3.23072%	1.28159%	0.07402%	0.00308%	0.00439%
Baa	0.00515%	0.13828%	6.06317%	89.02123%	4.20844%	0.50125%	0.04181%	0.02068%
Ba	0.00036%	0.01299%	0.28261%	8.25143%	85.30897%	5.27627%	0.28814%	0.57924%
B	0.00002%	0.00053%	0.01560%	0.22680%	4.73305%	88.89789%	3.44249%	2.68362%
Caa-C	0.00000%	0.00011%	0.00234%	0.04338%	0.50395%	17.96006%	68.42939%	13.06077%
D	0.00000%	0.00000%	0.00000%	0.00000%	0.00000%	0.00000%	0.00000%	100.00000%

4.2 INTERPOLATION AT THE OUTLOOK LEVEL

In order to do this, the steps involved are as follows:

- 1) Converting outlook level grades to numeric scale
- 2) Map the above mentioned numeric scale to transition matrix grades
- 3) Assign PD (=column D of regenerated 1 year transition matrix) to grades
- 4) Assign PD to outlook level by logit interpolation of grade level PD (from step 3)

The output of the algorithm is detailed in the table titled “**Outlook Level Interpolation**”.

The final column of the table (i.e. column 5) is the outlook level PD.

Table: Outlook Level Interpolation

1	2	3	4	5
Outlook Level Grades	Numeric Rank	Grade	1 year PD (equal to default rate column of the regenerated transition matrix)	Interpolated Outlook Level PD
Aaa	1	Aaa	0.00003%	0.00003%
Aa1	2			0.00010%
Aa2	3	Aa	0.00036%	0.00036%
Aa3	4			0.00083%
A1	5			0.00191%
A2	6	A	0.00439%	0.00439%
A3	7			0.00736%
Baa1	8			0.01233%
Baa2	9	Baa	0.02068%	0.02068%
Baa3	10			0.06289%
Ba1	11			0.19112%
Ba2	12	Ba	0.57924%	0.57924%
Ba3	13			0.96874%
B1	14			1.61588%
B2	15	B	2.68362%	2.68362%
B3	16			4.04266%
Caa1	17			6.04719%
Caa2	18			8.95291%
Caa3	19	Caa-C	13.06077%	13.06077%
Ca	20			18.66700%
C	21			25.96107%

5 APPENDIX-B : BANK PD

This section is meant for calibration of bank counterparty ratings. The methodology used here is exactly similar to that of the sovereign and hence is not discussed in detail. We only give here one year migration matrix and final outlook level PD

Table Migration Data: Exhibit 14 : Average One-Year Rating Transition Rates, 1983-2010: Financial Institutions (Debt Issuers Only)

From/To:	Aaa	Aa	A	Baa	Ba	B	CCC/C	D	NR
Aaa	86.32%	10.10%	0.08%	0.00%	0.00%	0.00%	0.00%	0.00%	3.50%
Aa	0.95%	85.22%	7.79%	0.30%	0.03%	0.01%	0.02%	0.04%	5.66%
A	0.10%	4.72%	84.22%	4.55%	0.51%	0.10%	0.03%	0.10%	5.66%
Baa	0.15%	0.68%	8.23%	78.96%	3.99%	1.01%	0.30%	0.31%	6.38%
Ba	0.00%	0.10%	0.82%	8.17%	71.83%	7.05%	1.08%	1.26%	9.68%
B	0.07%	0.10%	0.36%	0.65%	9.60%	65.99%	6.30%	4.07%	12.85%
Caa-C	0.00%	0.00%	0.00%	0.49%	1.18%	12.35%	45.51%	19.22%	21.25%

Source: Exhibit 14 : Defaults and Recoveries for Financial Institution Debt Issuers, 1983 – 2010, https://www.moodys.com/researchdocumentcontentpage.aspx?docid=PBC_128431

Table: Outlook Level Interpolation

1	2	3	4	5
Outlook Level Grades	Numeric Rank	Grade	1 year PD (equal to default rate column of the regenerated transition matrix)	Interpolated Outlook Level PD
Aaa	1	Aaa	0.00222%	0.00222%
Aa1	2			0.00971%
Aa2	3	Aa	0.04240%	0.04240%
Aa3	4			0.05755%
A1	5			0.07811%
A2	6	A	0.10600%	0.10600%
A3	7			0.15498%
Baa1	8			0.22653%
Baa2	9	Baa	0.33101%	0.33101%
Baa3	10			0.53541%
Ba1	11			0.86494%
Ba2	12	Ba	1.39444%	1.39444%
Ba3	13			2.09474%
B1	14			3.13555%
B2	15	B	4.66885%	4.66885%
B3	16			7.27334%
Caa1	17			11.16065%
Caa2	18			16.75025%
Caa3	19	Caa-C	24.37133%	24.37133%
Ca	20			34.04198%
C	21			45.25383%

Table: Simulation Results: Sovereign/Bank =10

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	578	5213	9
0.25	0.25	0.06	588	5316	9
0.25	0.25	0.08	594	5614	9
0.25	0.5	0.04	551	6547	12
0.25	0.5	0.06	606	6694	11
0.25	0.5	0.08	637	7177	11
0.25	0.75	0.04	606	9762	16
0.25	0.75	0.06	573	9839	17
0.25	0.75	0.08	589	10246	17
0.5	0.25	0.04	618	12022	19
0.5	0.25	0.06	564	12393	22
0.5	0.25	0.08	578	12565	22
0.5	0.5	0.04	624	13272	21
0.5	0.5	0.06	592	13745	23
0.5	0.5	0.08	636	14133	22
0.5	0.75	0.04	596	15971	27
0.5	0.75	0.06	590	16385	28
0.5	0.75	0.08	620	16958	27
0.75	0.25	0.04	580	21143	36
0.75	0.25	0.06	567	21541	38
0.75	0.25	0.08	570	22104	39
0.75	0.5	0.04	595	21697	36
0.75	0.5	0.06	607	22302	37
0.75	0.5	0.08	580	22879	39
0.75	0.75	0.04	558	22932	41
0.75	0.75	0.06	574	23025	40
0.75	0.75	0.08	577	23673	41

Table: Simulation Results: Sovereign/Bank =4

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	561	25175	45
0.25	0.25	0.06	609	25981	43
0.25	0.25	0.08	569	26777	47
0.25	0.5	0.04	634	29376	46
0.25	0.5	0.06	582	30012	52
0.25	0.5	0.08	525	31085	59
0.25	0.75	0.04	593	34305	58
0.25	0.75	0.06	585	35322	60
0.25	0.75	0.08	601	36046	60
0.5	0.25	0.04	587	40289	69
0.5	0.25	0.06	618	40904	66
0.5	0.25	0.08	564	42250	75
0.5	0.5	0.04	625	41824	67
0.5	0.5	0.06	627	42884	68
0.5	0.5	0.08	635	43662	69
0.5	0.75	0.04	557	43323	78
0.5	0.75	0.06	623	44363	71
0.5	0.75	0.08	546	44874	82
0.75	0.25	0.04	591	51053	86
0.75	0.25	0.06	536	52388	98
0.75	0.25	0.08	601	53190	89
0.75	0.5	0.04	656	50670	77
0.75	0.5	0.06	557	51607	93
0.75	0.5	0.08	580	52767	91
0.75	0.75	0.04	586	48596	83
0.75	0.75	0.06	588	49671	84
0.75	0.75	0.08	620	50551	82

Table: Simulation Results: Sovereign/Bank =2

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	660	71088	108
0.25	0.25	0.06	628	72943	116
0.25	0.25	0.08	588	75432	128
0.25	0.5	0.04	540	74456	138
0.25	0.5	0.06	574	76206	133
0.25	0.5	0.08	618	77993	126
0.25	0.75	0.04	623	72756	117
0.25	0.75	0.06	568	74065	130
0.25	0.75	0.08	562	75539	134
0.5	0.25	0.04	588	87879	149
0.5	0.25	0.06	641	89699	140
0.5	0.25	0.08	623	92067	148
0.5	0.5	0.04	598	86340	144
0.5	0.5	0.06	601	87814	146
0.5	0.5	0.08	570	89514	157
0.5	0.75	0.04	556	79471	143
0.5	0.75	0.06	587	80470	137
0.5	0.75	0.08	628	81987	131
0.75	0.25	0.04	567	91718	162
0.75	0.25	0.06	581	93171	160
0.75	0.25	0.08	598	93923	157
0.75	0.5	0.04	607	87525	144
0.75	0.5	0.06	559	88810	159
0.75	0.5	0.08	603	89898	149
0.75	0.75	0.04	630	80349	128
0.75	0.75	0.06	623	81653	131
0.75	0.75	0.08	606	82495	136

Table: Simulation Results: Sovereign/Bank =10: All banks have rating same as SBI

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	577	3000	5
0.25	0.25	0.06	615	3183	5
0.25	0.25	0.08	562	3155	6
0.25	0.5	0.04	570	3357	6
0.25	0.5	0.06	580	3368	6
0.25	0.5	0.08	563	3653	6
0.25	0.75	0.04	591	4697	8
0.25	0.75	0.06	573	4821	8
0.25	0.75	0.08	578	5018	9
0.5	0.25	0.04	613	6622	11
0.5	0.25	0.06	573	6780	12
0.5	0.25	0.08	591	6909	12
0.5	0.5	0.04	641	7153	11
0.5	0.5	0.06	570	7159	13
0.5	0.5	0.08	607	7359	12
0.5	0.75	0.04	600	8227	14
0.5	0.75	0.06	610	8214	13
0.5	0.75	0.08	592	8474	14
0.75	0.25	0.04	593	11265	19
0.75	0.25	0.06	588	11668	20
0.75	0.25	0.08	658	11947	18
0.75	0.5	0.04	626	11429	18
0.75	0.5	0.06	596	11772	20
0.75	0.5	0.08	583	12082	21
0.75	0.75	0.04	597	11788	20
0.75	0.75	0.06	612	11865	19
0.75	0.75	0.08	633	11901	19

Table: Simulation Results: Sovereign/Bank =4: All banks have rating same as SBI

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	595	12991	22
0.25	0.25	0.06	620	13124	21
0.25	0.25	0.08	594	13264	22
0.25	0.5	0.04	626	13561	22
0.25	0.5	0.06	584	13814	24
0.25	0.5	0.08	594	14076	24
0.25	0.75	0.04	604	14978	25
0.25	0.75	0.06	581	15025	26
0.25	0.75	0.08	637	15358	24
0.5	0.25	0.04	590	18405	31
0.5	0.25	0.06	592	18473	31
0.5	0.25	0.08	620	19183	31
0.5	0.5	0.04	612	18727	31
0.5	0.5	0.06	628	18823	30
0.5	0.5	0.08	606	19167	32
0.5	0.75	0.04	561	17917	32
0.5	0.75	0.06	541	18346	34
0.5	0.75	0.08	619	18604	30
0.75	0.25	0.04	568	22118	39
0.75	0.25	0.06	627	22570	36
0.75	0.25	0.08	606	22567	37
0.75	0.5	0.04	540	21306	39
0.75	0.5	0.06	601	21857	36
0.75	0.5	0.08	562	21903	39
0.75	0.75	0.04	643	19210	30
0.75	0.75	0.06	605	19517	32
0.75	0.75	0.08	623	19664	32

Table: Simulation Results: Sovereign/Bank =2: All banks have rating same as SBI

sovereign factor loading	bank factor loading	bank mean return	Standalone default count	Effective Default Count	Spillover Impact
0.25	0.25	0.04	607	27230	45
0.25	0.25	0.06	578	27477	48
0.25	0.25	0.08	620	28154	45
0.25	0.5	0.04	603	27217	45
0.25	0.5	0.06	601	27863	46
0.25	0.5	0.08	632	28518	45
0.25	0.75	0.04	594	25551	43
0.25	0.75	0.06	620	26089	42
0.25	0.75	0.08	592	26642	45
0.5	0.25	0.04	652	36066	55
0.5	0.25	0.06	643	36471	57
0.5	0.25	0.08	567	37329	66
0.5	0.5	0.04	587	33732	57
0.5	0.5	0.06	582	34246	59
0.5	0.5	0.08	640	34840	54
0.5	0.75	0.04	627	28416	45
0.5	0.75	0.06	605	29235	48
0.5	0.75	0.08	577	29198	51
0.75	0.25	0.04	595	36827	62
0.75	0.25	0.06	587	36830	63
0.75	0.25	0.08	576	37541	65
0.75	0.5	0.04	585	33200	57
0.75	0.5	0.06	573	33447	58
0.75	0.5	0.08	608	34482	57
0.75	0.75	0.04	615	27076	44
0.75	0.75	0.06	620	27166	44
0.75	0.75	0.08	620	27584	44

ⁱ World Development Indicators, 30 June 2017